Interactive Beat Annotation GUI – SOTA Draft II

Introduction

Music Information Retrieval (MIR) is a research field that develops computational methods to analyze, interpret, and retrieve musical information from audio, examples include: harmony, melody, rhythm, and structural elements. Within MIR, Beat Tracking (BT) focuses on the automatic extraction of musical beat information from recordings. Since the beat provides a fundamental temporal framework for music, BT is a critical component of many MIR systems.

“Many MIR systems first analyze the beats as the starting point, assume the results would be correct, and use them as a unit for further processing.” (Yamamoto, 2021)

The beat is typically defined as a regularly occurring pulse that underlies the temporal organization of a musical piece. However, the perception of the beat can vary across listeners. An objectively definitive beat can only be established if it is explicitly defined by the composer or agreed upon by performing musicians.

BT has been a research focus for decades. Existing algorithms perform reliably for music with steady, predictable tempos, such as Pop music. In contrast, music with expressive and/or fluctuating tempos remains challenging for automatic beat detection. Examples of music genres that are still difficult to assess are, classical, jazz or world music. These can make the process of beat tracking more complex and ambiguous.

For this reason SoTA BT autonomous algorithms are all machine learning based. And although there are a few that are self-supervised (Desblancs et al., 2023), these are usually genre specific or are bound to certain musical configurations. For the cases that SoTA BT still can’t solve, human-made beat annotation (or revisions) are still the most reliable.

Although there has been a lot of effort in advancing autonomous BT solutions, the same can not be said for human focused BT tools where it seems that they have been stayed largely unchanged for years now.

Research Questions (Tenho de reescreve-las)

With this, there are a couple of research question that summarize the problems that we intend to tackle:

1. How can we improve the accuracy and efficiency of contemporary beat annotation workflows?
2. How can we make an effective music score representation, that can be aligned to show notes in their absolute time positions?
3. How can we make such score align with a spectogram view of the recording?

SOTA (está demasiado desorganizado)

Although Beat Tracking is a research filed that can go as far back as 1995 at least [8], the tools being used for this task stayed largely the same for a years. Not only that, software that allows for Beat Tracking, such as Audacity and Sonic Visualizer (SV) are programs that are designed for a multitude of other audio related tasks. When it comes to Beat Tracking optimized software, there are few options if none. In 2023 Yamamoto proposed a Human-In-the-Loop approach for BT [2], the problem is, although this paper seems promising, the software developed for this paper is closed-sourced, property of Yamaha Corp. and thus, it is not likely to be publicly accessible.

There are lots of software for all sorts of MIR tasks. Due to it´s open source nature, SV has enabled a lot of different MIR task specific software to be developed such as Piano Precision [7]: for score to audio "follow along". There have been attempts to make web-based technology for enabling active listening and collaboration on various MIR tasks, examples include Songle (2011) [5] and Dezrann (2025) [6]. But crucially, none are BT focused. |

Sonic Visualizer is still the go-to software for scientific and MIR purposed audio analysis. However when it comes to BT, SV is outdated due to it's inefficient BT labeling process, that is still very much manual and cumbersome [2]. SV also is heavily reliant on VAMP plug-ins, modern SoTA BT algorithms are complex ML applications that may be difficult to implement into VAMP plug-ins and/or there is no direct or valued incentive for making these algorithms available as VAMP plug-ins.

Time Instances inside SV (name of the tool in SV for BT), can be exported and imported as simple .txt files. Thus, for one to benefit from a SoTA BT algorithm, one needs to follow a few steps: First, one needs to have access the algorithm in itself which might involve not only browsing, but also probably, accessing some publicly available source-code repository. Secondly, one needs to know how to compile and run the program and export the retrieved data into a .txt file that SV can read. Besides this, we also need to take into account that to accomplish these steps, one needs at least, to be minimally literate/experienced with programming.

One problem with this, is that other users who may not be experienced in programming or literate in MIR research will rely on the most easily accessible BT algorithm inside SV, and these will probably be Beat Trackers like: BeatRoot (2001) [11], Tempo and Beat Tracker - Queen Mary University of London (2009) [12] and IBT – INESC Beat Tracker (2010) [13] among others. The reason being that, although they don't come pre-installed with SV, they can be added via a SV native feature that allows to browse and install VAMP plugins without exiting the program (referring to the "Find Transform" SV feature). Crucially it is important to note that, as of the writing of this paper, you will not find any SoTA BT using this SV built in feature, in fact one will only find outdated BT that are at least one decade old.

For music in which autonomous SoTA BT systems aren't efficient enough, relaying on human made beat annotations still might be preferable to the annotator:

"A common method to create beat annotations for music recordings is to let a human annotator tap along with them." [10]

The problem with this, is that these annotations will probably not be accurate enough even if the user preforming them is an experienced musician. There are multiple reasons that stem from perception, human motor noise, jitter and system latency [10]. Thus when tapping, there is the need to shift marked beats in order to make them accurate:

"This shifting operation is particularly relevant when correcting tapped beats, which can be subject to human motor noise as well as jitter and latency during acquisition." [9]

One of the most common BT assessment mistakes are the Double/Half time errors, in which the BT algorithm might mark the beat in orders of integer multiplication (or division) from the actually correct beat. Another common mistake might be inducing multiple closely sequenced markings in the place where only one should be, or the contrary, omitting/skipping beats. For correcting these type of mistakes, current Beat Tracking software only allow correcting beats one-by-one. A simple workaround for this mistake would be to develop a feature in which the user could selects a time region of beats and perform an operation such as: 'In this region, that start at xx:xx and ends at xx:xx , the tempo is a steady X bpm with Y/Z timeSignature' the program would then shift/eliminate/highlight marked beats that deviate from the "user statement".

Something that is not very well documented is the relevance and importance of following along a music score when beat tapping. Besides helping the annotator follow along more accurately the recordging, a score might notice the annotator in advance of any bpm or time signature changes. Besides that, when using a Spectrogram-Based view of the recording, one can distinctively align a specific onset to a concrete and exact moment in the recording timeline, this is essential to know if a beat is accurately assessed or not.

Piano Precision [7] is an example of an SV fork that was designed for displaying musical scores alongside audio recordings in a unified graphical interface. This program works surprisingly well for Solo Piano pieces. Now there are two question that arrive from this:

One is that, by following along the music sheet, Beat Tracking autonomously should be as straight forward as making the program align a beat marking with the respective note onsets that are marked in the score.

The other would be to integrate this idea of Audio-to-Score alignment with a Spectral View of the recording. In PianoPrecision this is done via a side-window that displays the a single A4 page of the score, one at a time. When the user hits play, a darkened highlight displays the current playback position in the score, while in another window a spectral view of the audio is is following along the playback position horizontally. This allows for a musically natural follow-along of the score, mimicking how a musical score is actually read. However, for audio analysis and visualization purposes, it could be beneficial for the score to be aligned and viewed along side the playback position, meaning, horizontally and with note onsets that are aligned with their respective absolute position in the timeline. For BT this could allow the user to "grab a beat" and position it on the it's corresponding position on top of the spectogram representation. Also this could be beneficial for more easily detecting beats that should be shifted.

Yamamoto's Human-In-the-Loop proposal [2] seems to be a step in the right direction for a BT UI. However,

a flaw with his paper is, that it fails to mention how a spectrograms can be beneficial as opposed to the more conventional waveform representations, for the act of beat tracking. While a waveform representation can be useful for identifying transients and general differences in energy along audio file. Spectograms can besides that, represent more accurately information related with spectral energy in time. A clear example of this would be: In a waveform, one can distinguish a ff section of from a pp in a recording. In a spectogram, one can not only distinguish these, but also understand if a particular section would be chord hit, or an arpeggio, along side noticing where these note onsets would be located pitch wise.

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